

emission of radiation)

LASER [Light amplification by stimulated

classmate

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Page

A device which produces a light beam having high intensity, monochromatic, coherent and unidirectional nature is LASER.

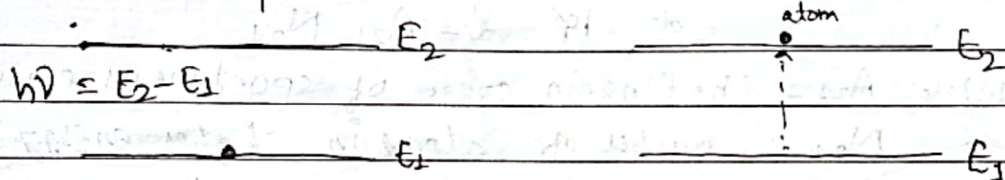
The basic principle of LASER is quantum theory of radiation which was worked by the scientist Townes and Schawlow in 1958.

According to the quantum theory of radiation, there are three types of absorption and or emission between the two energy levels E_1 and E_2 such that $E_2 > E_1$.

1. Induced absorption:-

When an atom is initially in the lower energy level and a photon of energy $h\nu = E_2 - E_1$ is incident towards the atom, the atom itself absorbs the incident photon and moves towards the higher energy levels state.

Then this type of absorption is known as induced absorption.



According to Einstein the rate of induced absorption is directly proportional to the no. of atoms in the lower energy levels and energy density of incident photon.

i.e. $\frac{dN_{12}}{dt} \propto N_{12}\psi$

$$\Rightarrow \left[\frac{dN_{12}}{dt} \right]_{ab} = B_{12} N_{12} \psi$$

Where B_{12} = proportionality constant known as Einstein's coefficient of induced absorption

N_{12} = no. of atoms in lower energy level

ψ = energy density of incident photon

b) Spontaneous emission:-

When an atom is initially in the higher energy state, this atom remains there for a short interval of time (i.e. about 10^{-8} sec). After this time the atom returns to the lower energy level by emitting a photon having energy $h\nu = E_2 - E_1$. This type of emission is called the spontaneous emission.

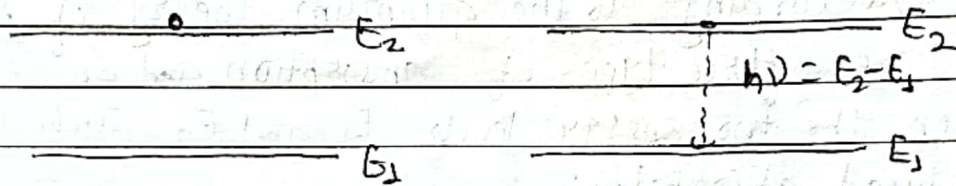


Fig:- Spontaneous emission.

According to Einstein, the rate of spontaneous emission is directly proportional to the number of atoms in the higher energy level.

$$\text{i.e. } \frac{dN_2}{dt} \bigg|_{sp} = -A_{21} N_2$$

Where, A_{21} = the Einstein coeff. of spontaneous emission.

N_2 = number of atoms in higher energy level.

c) Stimulated emission:-

Under certain conditions, it is possible to force an atom in the higher energy level to emit an additional photon similar to incident photon. For this when an atom is initially in the higher energy level, then a photon is projected towards the atom. The atom in the higher energy level interacts with the incident photon and emits an additional photon similar to incident photon. This type of emission is known as stimulated emission. The incident photon together with the emitted photon move in the same direction and trigger another emission. This process is multiplied. As a result of

this, a beam of photons having monochromatic, ~~chaos~~ coherent, unidirectional and high intensity is produced, and is known as laser.

According to the Einstein the rate of stimulated emission is directly proportional to the number of atoms in the higher energy level and energy density of the incident photons.

$$\text{i.e. } \frac{dN_2}{dt} \propto N_2 \psi$$

$$\frac{dN_2}{dt} = B_{21} N_2 \psi$$

where, B_{21} = proportionality constant known as the Einstein coefficient of stimulated emission

N_2 = no. of atoms in higher energy level.

ψ = energy density of incident photon.

Active medium :- (mediums)

Different media i.e. solid, liquid and gas are the heart of the laser; when these mediums contain ions, atoms or molecules. If these ions atoms or molecules are capable of undergoing transition from higher energy level to lower energy level in the radioactive manner, then these atoms ions and molecules are called active species or active center. Thus the corresponding mediums are known as active mediums.

Population inversion and pumping :-

(ϕ)

According M.B. (Maxwell-Boltzmann) statistics the no. of atoms in the lower energy level is always greater than the number of atoms in higher energy level at ordinary conditions.

$$\text{i.e. } N_2 = N_1 e^{-\frac{E_2 - E_1}{kT}}$$

$K = \text{Boltzmann's constant}$

$T = \text{absolute temperature.}$

At that condition induced absorption dominates the stimulate emissions, but under certain condition it is possible to make no. of atoms in the higher energy level greater than the number of atoms in the lower energy level. At that time stimulated emission dominates the induced absorption, therefore this condition at which the number of atoms in the higher energy level is greater than the number of atoms in lower energy level is called population inversion. This is essential for the LASER action.

The method of obtaining population inversion is called pumping. There are four types of pumps.

a) Optical pumping :- Method of obtaining the population inversion by using the light of suitable wavelength is called optical pumping.

b) Electric pumping :- Method of obtaining the population inversion by using the electric current is called electric pumping.

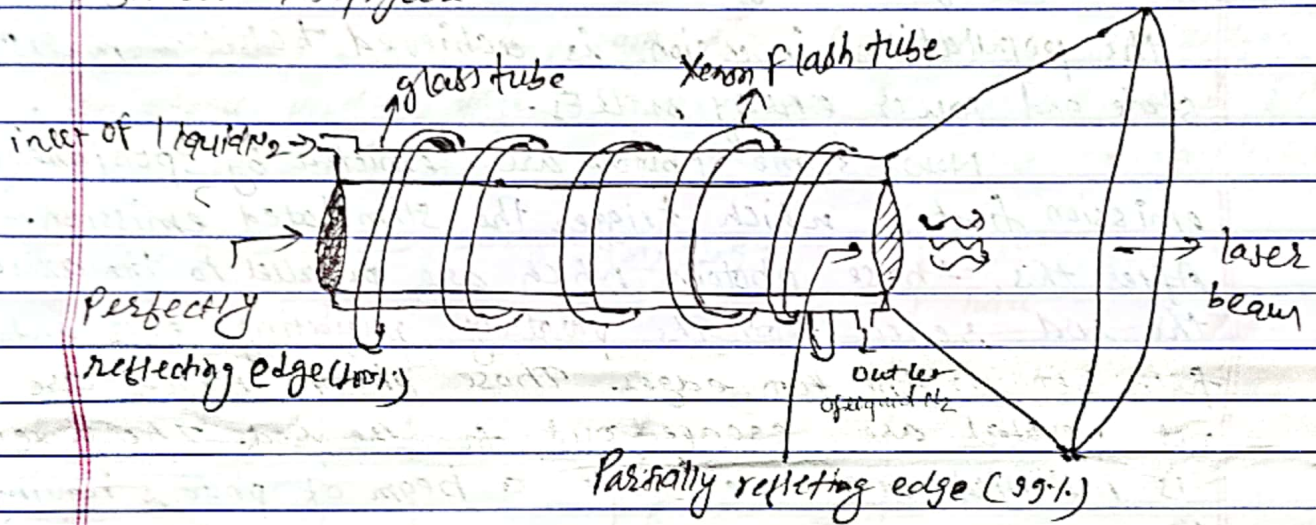
c) Heat pumping :- - - - - by using heat - - - - .

d) Chemical pumping :- - - - - by using chemical reaction - - - - is called chemical pumping.

Ruby Laser (Civil)

Ruby is a naturally occurring precious crystal of aluminium oxide (Al_2O_3). The ruby for laser action is doped with 0.05% chromium oxide (Cr_2O_3), where some aluminium ions are replaced by chromium ions which act as active centre.

The schematic diagram of ruby laser is shown in figure.



The ruby laser consists of a ruby rod having two parallel and flat edges. The rod is kept inside a glass tube mounted by a spiral xenon flash tube which acts as pumping system in the ruby laser as shown in the figure.

When light flash, having wavelength 550nm , is given to the ruby rod from the xenon flash tube, the chromium ions are excited to the higher energy level E_3 . Here the life time of the ions is very short (10^{-8}sec), so the ions undergo non-radiative transition from E_3 to meta-stable state E_2 as shown in the figure.

The life time at metastable state is comparatively larger ($3 \times 10^{-3}\text{sec}$). Therefore the

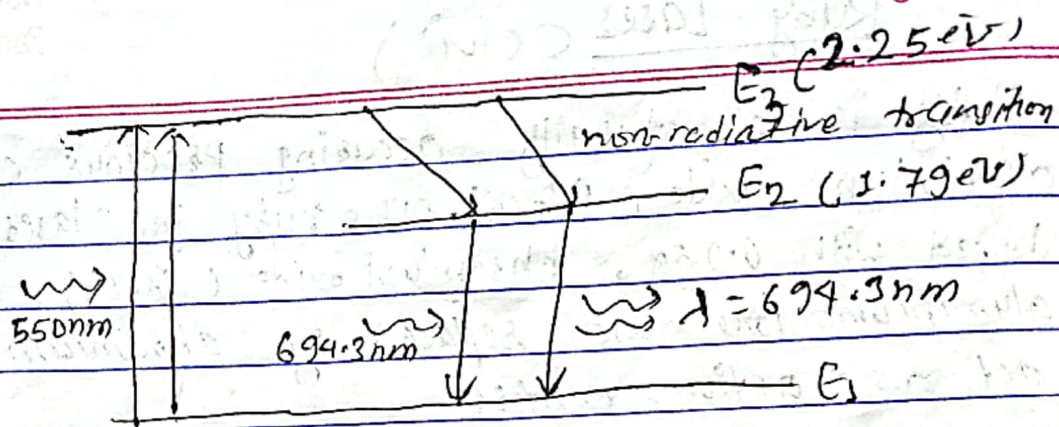


Fig.: Energy level diagram of Cr^{3+} ion.

the population inversion is achieved between metastable state and lower energy state E_1 .

Now some photons are "emitted" by spontaneous emission first, which trigger the stimulated emission. After this, those photons which are parallel to the axis of the rod reflect from the partially reflecting end and forth between the two edges. Those photons which are not parallel are escaped out from the rod. The process is multiplied and as a result, a beam of photons having high intensity, monochromatic, coherent and unidirectional property is emitted from partially silvered end of the rod. This beam is known as laser beam.

Computer (B.E.) He-Ne gas laser:-

The schematic diagram of He-Ne laser is as shown in figure.

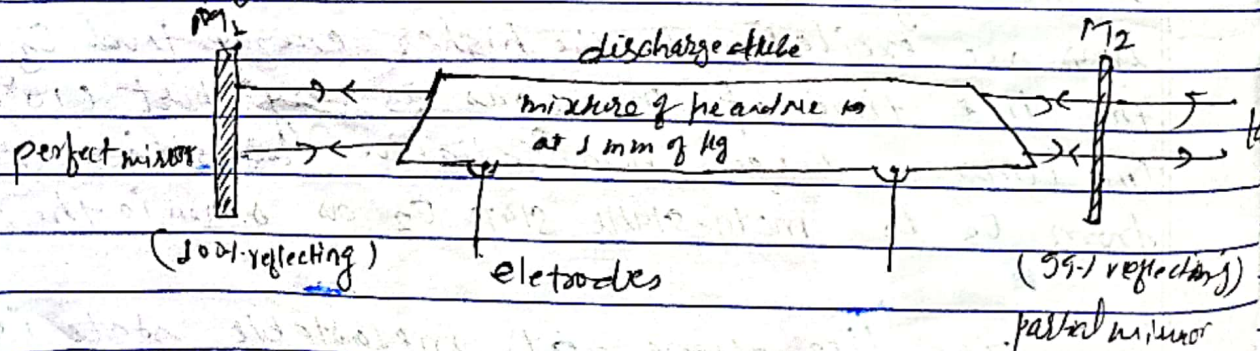


Fig.: He-Ne laser

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He-Ne laser consists of a discharge tube, whose ends are cut in Brewster's angle, in which a mixture of He-Ne gas in the ratio of 5:1 is kept at a pressure of 1 mm of Hg. Also it consists of two electrodes which act as a pumping system. Besides it consists of ~~two~~ two mirrors M_1 and M_2 in which M_1 is perfect mirror and M_2 is partial mirror.

When dc current is passed into mixture of gases through electrodes, some gas atoms undergo ionization and hence produce free electrons. These free electrons collide with the He-Ne atoms and excite them to different energy levels.

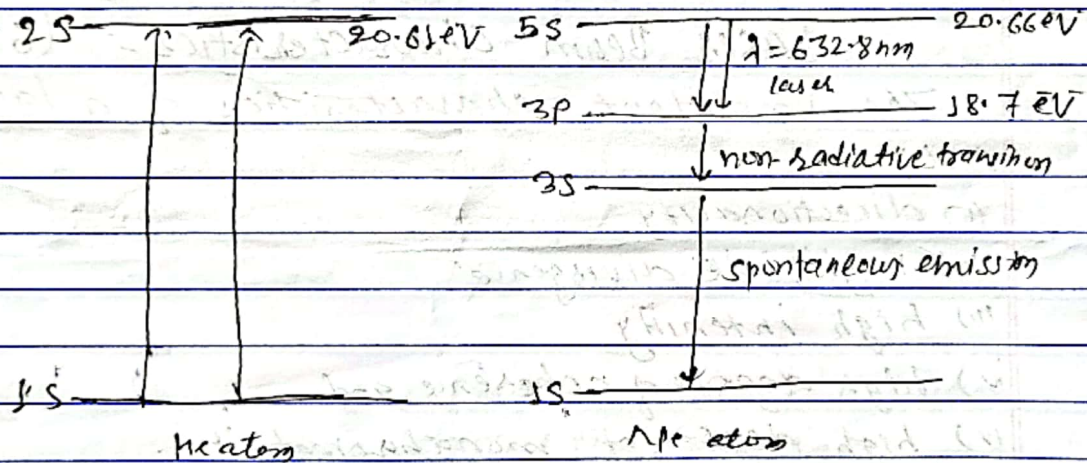


Fig: Energy level diagram of He and Ne atoms.

Here the excitation efficiency of He is much more than that of Neon, and also energy of 2s level of Helium is approximately equal to the 5s level of Neon. Therefore the energy from 2s level of Helium undergoes transition transformation to the 5s level of Neon. When they collide with each other, thus population inversion is achieved between 5s and 3p of Neon atom, but He-atom only helps in the pumping system.

In this case some photons are excited by

the spontaneous emission which then triggers the stimulated emission. Those photons which are parallel to the axis of tube reflect back and forth between the mirrors, but those not parallel to ~~the~~ are eliminated. This process is repeated in the geometrical manner. As a result of this, a beam of photons having ~~a~~ ~~mon~~ properties of monochromatic, coherent, unidirectional and highly intensive is produced. This beam comes out from partial mirror as laser beam.

Laser Beam Characteristics - (831-832)

The important characteristics of a laser beam are:

- i) directionality
- ii) negligible divergence
- iii) high intensity
- iv) high degree of coherence, and
- v) high ~~degree~~ of monochromaticity.

Applications:

LASER is an interesting and most exciting inventions of 20th century. It is used for the following works:

2) For the material works: (a) cutting (b) healing
(c) drilling (d) penetration (e) welding

3) For medical works:

g) To perform the painless and bloodless surgery.

h) To cure the tumor, ulcers and bone fracture

i) For cutting blood vessels and welding them.

3) For communication works

i) to set the communication between the satellites
satellites, satellite and station submarines, or
submarines and station etc.

ii) It is used in ~~the~~ optical fibre.

iii) It is also used in ~~the~~ computers.

4) LASER is used to detect the air pollution
in the atmosphere, for this LIDAR is used
(LIDAR $\rightarrow$ light detection and ranging)

5) Laser is used in holography

6) Laser is used in to separate the isotopes

7) Laser is used to ~~detect~~ forecast the weather
condition

8) Laser is used to check the function of
Michelson's interferometer.

31/12/2022 07:53